

Analyzing Ellipses: Standard Form, Vertices & Foci

Precalculus / Analytic Geometry Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Identify the center, vertices, and orientation of an ellipse from its graph or equation
- Write the standard form equation of an ellipse given key features
- Find the foci (foci) of an ellipse using the relationship between a, b, and c

Problems

1. Identify the center and state whether the major axis is along the x-axis or y-axis for the ellipse with the equation below.

$$\frac{x^2}{9} + \frac{y^2}{25} = 1$$

2. For the ellipse given below, find the values of a and b, then list all four vertices.

$$\frac{x^2}{49} + \frac{y^2}{16} = 1$$

3. An ellipse is centered at the origin with vertices at (0, 6), (0, -6), (4, 0), and (-4, 0). Write the standard form equation of this ellipse.

4. Rewrite the equation below in standard form by dividing both sides by 225, then identify a^2 and b^2 .

$$25x^2 + 9y^2 = 225$$

5. Find the foci of the ellipse whose equation is given below. Use the relationship $c^2 = b^2 - a^2$ where b^2 is the larger denominator.

$$\frac{x^2}{9} + \frac{y^2}{25} = 1$$



6. Find the foci of the ellipse with the equation below.

$$\frac{x^2}{49} + \frac{y^2}{24} = 1$$

7. Write the standard form equation of an ellipse centered at (3, -2) with a = 5 along the x-direction and b = 3 along the y-direction.

8. Convert the equation below to standard form by completing the square, then identify the center, a, and b.

$$4x^2 + 9y^2 - 16x + 36y + 16 = 0$$

9. An ellipse is centered at (-1, 4) with foci at (-1, 4 + 2√5) and (-1, 4 - 2√5), and b = 3 (minor semi-axis along the x-direction). Find a and write the standard equation.

10. Find all key features — center, a, b, c, vertices along both axes, and both foci — for the ellipse with the equation below.

$$\frac{(x + 2)^2}{36} + \frac{(y - 5)^2}{100} = 1$$



Analyzing Ellipses: Standard Form, Vertices & Foci — Answer Key

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Answer Key

1. Answer: Center: (0, 0); major axis along the y-axis

- The equation is already in standard form with $h = 0$ and $k = 0$, so the center is $(0, 0)$.
- Compare denominators: $25 > 9$, and 25 is under y^2 , so the major axis is along the y-axis.

2. Answer: $a = 7$, $b = 4$; vertices: (7, 0), (−7, 0), (0, 4), (0, −4)

- Since $49 > 16$ and 49 is under x^2 , the major axis is along the x-axis, so $a^2 = 49 \rightarrow a = 7$.
- $b^2 = 16 \rightarrow b = 4$.
- Major-axis vertices: $(\pm 7, 0)$; minor-axis vertices: $(0, \pm 4)$.

3. Answer: $x^2/16 + y^2/36 = 1$

- The vertices along the y-axis are $(0, \pm 6)$, so $b = 6$ (major axis along y-axis in this orientation).
- The vertices along the x-axis are $(\pm 4, 0)$, so $a = 4$.
- Standard form: $x^2/16 + y^2/36 = 1$.

4. Answer: $x^2/9 + y^2/25 = 1$; $a^2 = 9$, $b^2 = 25$

- Divide every term by 225: $25x^2/225 + 9y^2/225 = 1$.
- Simplify: $x^2/9 + y^2/25 = 1$.
- Therefore $a^2 = 9$ (under x^2) and $b^2 = 25$ (under y^2).

5. Answer: Foci: (0, 4) and (0, −4)

- Since $25 > 9$, the major axis is along the y-axis. $b^2 = 25$, $a^2 = 9$.
- $c^2 = b^2 - a^2 = 25 - 9 = 16$, so $c = 4$.
- Foci lie along the y-axis: $(0, 4)$ and $(0, -4)$.

6. Answer: Foci: (5, 0) and (−5, 0)

- Since $49 > 24$, the major axis is along the x-axis. $a^2 = 49$, $b^2 = 24$.
- $c^2 = a^2 - b^2 = 49 - 24 = 25$, so $c = 5$.
- Foci lie along the x-axis: $(5, 0)$ and $(-5, 0)$.

7. Answer: $(x - 3)^2/25 + (y + 2)^2/9 = 1$

- Center is $(h, k) = (3, -2)$, $a = 5$ (along x-axis), $b = 3$ (along y-axis).
- Substitute into standard form: $(x - 3)^2/a^2 + (y - (-2))^2/b^2 = 1$.
- Result: $(x - 3)^2/25 + (y + 2)^2/9 = 1$.

8. Answer: $(x - 2)^2/9 + (y + 2)^2/4 = 1$; center (2, −2), $a = 3$, $b = 2$

- Group x and y terms: $(4x^2 - 16x) + (9y^2 + 36y) = -16$.
- Factor: $4(x^2 - 4x) + 9(y^2 + 4y) = -16$.



- Complete the square: $4(x - 2)^2 - 16 + 9(y + 2)^2 - 36 = -16 \rightarrow 4(x-2)^2 + 9(y+2)^2 = 36$.
- Divide by 36: $(x-2)^2/9 + (y+2)^2/4 = 1$. Center $(2, -2)$, $a = 3$, $b = 2$.

9. Answer: $a = 7$; $(x + 1)^2/9 + (y - 4)^2/49 = 1$

- The foci are along the y-direction, so the major axis is vertical. $c = 2\sqrt{5}$, $b = 3$.
- Use $c^2 = a^2 - b^2$: $(2\sqrt{5})^2 = a^2 - 9 \rightarrow 20 = a^2 - 9 \rightarrow a^2 = 29$. Wait — re-check: $c = 2\sqrt{5}$ means $c^2 = 20$, $b = 3$ means $b^2 = 9$, so $a^2 = 20 + 9 = 29$ and $a = \sqrt{29}$.
- Correction note: $a^2 = 29$, $a = \sqrt{29}$.
- Standard form: $(x + 1)^2/9 + (y - 4)^2/29 = 1$.

10. Answer: Center $(-2, 5)$; $a = 6$ (x), $b = 10$ (y); $c = 8$; vertices: $(4,5), (-8,5), (-2,15), (-2,-5)$; foci: $(-2,13), (-2,-3)$

- Center: $h = -2$, $k = 5 \rightarrow (-2, 5)$.
- Since $100 > 36$, major axis is along y-direction. $b^2 = 100 \rightarrow b = 10$; $a^2 = 36 \rightarrow a = 6$.
- $c^2 = b^2 - a^2 = 100 - 36 = 64 \rightarrow c = 8$.
- Major-axis vertices (along y): $(-2, 5 \pm 10) = (-2, 15)$ and $(-2, -5)$.
- Minor-axis vertices (along x): $(-2 \pm 6, 5) = (4, 5)$ and $(-8, 5)$.
- Foci (along y): $(-2, 5 \pm 8) = (-2, 13)$ and $(-2, -3)$.

